Summer paper II title

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1. Model Formulation

Consider an infinite-horizon discrete-time setting, indexed by $t \in \mathbb{N}^+ = \{1, 2, \dots\}$. The environment consists of a single user and two sellers, denoted by A and B.

At each period t, the user selects a seller, denoted by the action $y_t \in \{A, B\}$. The user's realized utility in period t is a binary random variable $u_t \in \{0, 1\}$.

If the user selects Seller B ($y_t = B$), they are offered a single, transparent product. The realized utility is drawn from a Bernoulli distribution with a strictly known parameter $p_0 \in (0, 1)$:

$$\mathbb{P}(u_t = 1 \mid y_t = B) = p_0. \tag{1}$$

If the user selects Seller A ($y_t = A$), Seller A dynamically assigns a product from a finite set $\mathcal{X} = \{1, 2\}$. Let $x_t \in \mathcal{X}$ denote Seller A's choice. The assigned product determines both the probability of a successful outcome for the user and the cost incurred by Seller A. Specifically, product $i \in \mathcal{X}$ yields $u_t = 1$ with probability $p_i \in (0, 1)$ and imposes a cost $c_i \in \mathbb{R}_+$ on Seller A.

$$\mathbb{P}(u_t = 1 \mid y_t = A, x_t = i) = p_i. \tag{2}$$

Without loss of generality, we assume $c_1 < c_2$. Furthermore, we assume that whenever Seller A is chosen ($y_t = A$), they receive a fixed revenue $R > 0$. The immediate profit for Seller A in period t is thus:

$$\pi_t^A = \begin{cases} R - c_{x_t} & \text{if } y_t = A, \\ 0 & \text{if } y_t = B. \end{cases} \tag{3}$$

2. Information Structure and Learning Dynamics

The model's core tension stems from an information asymmetry between the user and Seller A.

ASSUMPTION 1 **(User's Misspecified Belief)**. *The user observes the realized utility u_t but cannot observe the specific product $x_t \in \mathcal{X}$ assigned by Seller A. Consequently, the user operates under the misspecified belief that Seller A offers a single product characterized by an unknown success probability $\theta \in [0,1]$.*

The user learns θ through Bayesian updating. Let the user's prior belief over θ at $t = 1$ be uniformly distributed, given by $\theta \sim \mathrm{Beta}(1,1)$. Let S_t and F_t denote the cumulative number of successes and failures, respectively, generated by Seller A up to the beginning of period t:

$$S_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 1\}, \tag{4}$$

$$F_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 0\}, \tag{5}$$

The user's posterior belief at period t follows a Beta distribution:

$$\theta \mid S_t, F_t \sim \mathrm{Beta}(1 + S_t, 1 + F_t). \tag{6}$$

The user follows a Thompson Sampling heuristic for their decision policy. At the beginning of period t, the user samples an index $\tilde{\theta}_t \sim \mathrm{Beta}(1 + S_t, 1 + F_t)$. Because the expected utility of Seller B (p_0) is perfectly known, the user selects Seller A if and only if the sampled index exceeds p_0:

$$y_t = \begin{cases} A & \text{if } \tilde{\theta}_t \geq p_0, \\ B & \text{if } \tilde{\theta}_t < p_0. \end{cases} \tag{7}$$

3. Seller A's Optimization Problem

Seller A observes the sequence of the user's choices and the realized utilities. Therefore, Seller A perfectly tracks S_t and F_t, thereby possessing complete information regarding the user's internal belief state.

Let the state of the system from Seller A's perspective be defined by the user's sufficient statistics, $\boldsymbol{s}_t = (S_t, F_t) \in \mathbb{N}_0 \times \mathbb{N}_0$. Given state $\boldsymbol{s}_t$, the probability that the user chooses Seller A in period t, denoted $\rho(S_t, F_t)$, is the complementary cumulative distribution function (CCDF) of the Beta distribution evaluated at p_0:

$$\rho(S_t, F_t) \triangleq \mathbb{P}(\tilde{\theta}_t \geq p_0 \mid S_t, F_t) = \int_{p_0}^{1} \frac{\theta^{S_t}(1-\theta)^{F_t}}{B(1+S_t, 1+F_t)} \, d\theta, \tag{8}$$

where $B(\cdot,\cdot)$ is the Beta function.

Seller A faces a Markov Decision Process (MDP) where they must balance immediate cost minimization against the strategic manipulation of the user's belief state to ensure future engagement. Let $\gamma \in (0,1)$ denote Seller A's discount factor.

Problem 1 (Seller A's MDP). Seller A seeks a stationary policy $\pi_A : \mathbb{N}_0 \times \mathbb{N}_0 \to \mathcal{X}$ that maximizes the expected discounted cumulative profit:

$$\max_{\pi_A} \mathbb{E}_{\pi_A} \left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbb{1}\{y_t = A\}(R - c_{x_t}) \right]. \tag{9}$$

Let $V(S,F)$ denote the optimal value function for Seller A at state (S,F). The state transitions occur only when the user selects Seller A. If $y_t = B$, the state remains (S,F) and no profit is accrued, though time advances. For each product $x \in \{1,2\}$, define the product-specific action value

$$Q_x(S,F) \triangleq (R - c_x) + \gamma \big[p_x V(S+1, F) + (1-p_x) V(S, F+1) \big]. \tag{10}$$

This quantity is Seller A's expected payoff from using product x, conditional on the user choosing Seller A in the current period. The Bellman optimality equation is therefore given by:

$$V(S,F) = \big[1 - \rho(S,F)\big] \gamma V(S,F) + \rho(S,F) \max_{x \in \{1,2\}} Q_x(S,F). \tag{11}$$

By algebraic manipulation, we can isolate $V(S,F)$ to yield a computationally tractable form that resolves the self-referential term arising from periods where Seller A is not chosen:

$$V(S,F) = \frac{\rho(S,F)}{1 - \gamma + \gamma \rho(S,F)} \max_{x \in \{1,2\}} Q_x(S,F). \tag{12}$$

Seller A's best response at state (S,F) is any product attaining this maximum. In particular, product 2 is optimal whenever $Q_2(S,F) \geq Q_1(S,F)$.